Title of the Article

Khaled Surname*[1], Second Author[1], and Third Author[2]

[1]Lab-STICC, UMR CNRS 6285, ENSTA, Brest, France
[2]Other Institution, City, Country

Abstract

Write a concise summary (150–250 words): context, problem, proposed method, main results, and significance.

Keywords: keyword one; keyword two; keyword three; keyword four

1 Introduction

Context and motivation. Related work can be cited like this [**example2024**]. State the contributions clearly:

- first contribution;
- second contribution;
- third contribution.

The rest of the paper is organized as follows: **??** presents the method, **??** the results, and **??** concludes.

2 Geometry and signal model

Every channel used in this paper is, to first order, a combination of two fixed functions of one normalised coordinate. this section sets the geometry, derives that coordinate and lists what the platform actually measures.

2.1 Survey Geometry

The vehicle moves at speed $v = 1.0\,\mathrm{m\,s^{-1}}$ along straight legs that cross a straight cable at an incidence θ (Figure 1a). Two tri-axial fluxgates are mounted at the water line on the vehicle's lateral axis, UNO on the port side and DUO on the starboard side, separated by the baseline $b = 1\,\mathrm{m}$. For each sensor we use a local CPA frame: the origin is the point of its track directly

*Corresponding author: email@ensta.fr

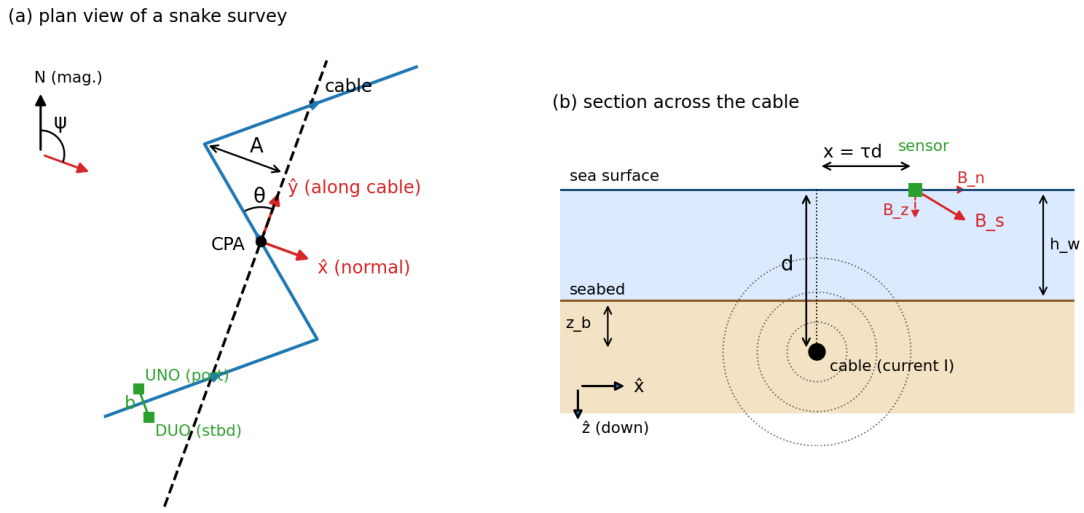


Figure 1: Survey geometry. (a) Plan view of a snake survey: cable (dashed), legs crossing it at incidence θ with amplitude A, CPA frame ($\hat{\boldsymbol{x}}$ normal, $\hat{\boldsymbol{y}}$ along the cable), the two sensors UNO and DUO separated by the baseline b, and the angle ψ from magnetic north to the cable normal. (b) Section across the cable: the sensor at the surface, the cable at standoff d below it (burial depth z_b below a seabed at water depth h_w), the normal coordinate $x = \tau d$, and the azimuthal cable field $\boldsymbol{B}_s$ with its normal and vertical components B_n and B_z.

above the cable, $\hat{\boldsymbol{x}}$ is the horizontal normal to the cable, $\hat{\boldsymbol{y}}$ points along the cable and $\hat{\boldsymbol{z}}$ points down (Figure 1b). The cable then lies at $x = 0$, $z = d$, where d is the standoff.

A sensor that crosses the cable at time t_c has horizontal distance $x = v\,(t - t_c)\sin\theta$ from the cable at time t. Every field quantity below depends on x only through the ratio x/d, so we use the normalised coordinate

$$\tau = \frac{x}{d} = \frac{v\,(t - t_c)\sin\theta}{d} = \frac{t - t_c}{T_s}, \qquad T_s = \frac{d}{v\sin\theta}. \tag{1}$$

τ counts CPA distances, and T_s is the time the sensor takes to move one CPA distance across the cable; at $\theta = 50°$ and $v = 1\,\mathrm{m\,s^{-1}}$ it is 2.6 s for $d = 2\,\mathrm{m}$ and 5.2 s for $d = 4\,\mathrm{m}$. T_s is the time scale of every matched template, and we call it the template scale. A leg that starts and ends at the lateral distance A from the cable covers $|x| \le A$, i.e. $|\tau| \le K$ with

$$K = \frac{L}{d}, \qquad L = A \text{ for a snake leg.} \tag{2}$$

L is the half-length of the pass measured across the cable. With the snake amplitude $A = 6\,\mathrm{m}$ used in closed loop and $d = 2\,\mathrm{m}$ to $4\,\mathrm{m}$, $K = 3$–1.5: a snake leg sees only a few CPA distances on each side of the crossing.

2.2 Cable Field and Cable Orthonormal Basis Functions

We follow the steps of [1]. The cable is modeled as an infinitely long straight conductor carrying a DC current I, with the return spread through the sea far from the sensor. Its field at the

sensor is azimuthal about the cable (Appendix A.1):

$$\mathbf{B}_s(\tau) = K_p \left(\frac{1}{1+\tau^2},\ 0,\ \frac{\tau}{1+\tau^2} \right)^T = (B_n, B_y, B_z)^T, \qquad K_p = \frac{\mu_0 I}{2\pi d}. \tag{3}$$

The components are given in the CPA frame: B_n is along the normal $\hat{x}$, $B_y = 0$ is along the cable, and B_z is downwards. K_p is the field magnitude at CPA, d is the standoff distance and $\tau = x/d$ is the normalized along-track position.

The infinite-line model holds when the cable is straight over several standoffs on each side of the crossing. The simulations instead use the exact field of a polyline made of 2 m segments.

Inserting (3) into the first-order term of the total-field anomaly gives a sum of two fixed functions of τ. Normalised to unit energy, they are the cable OBF of [1]:

$$\Phi_1(\tau) = \sqrt{\frac{2}{\pi}} \frac{1}{1+\tau^2}, \qquad \Phi_2(\tau) = \sqrt{\frac{2}{\pi}} \frac{\tau}{1+\tau^2}. \tag{4}$$

Φ_1 is even and Φ_2 is odd; both have unit energy over the real line, and they are orthogonal(Appendix A.3). The total-field anomaly is then

$$S(\tau) = A_1 \Phi_1(\tau) + A_2 \Phi_2(\tau), \qquad A_1 = \sqrt{\frac{\pi}{2}}\, K_p \cos\delta \cos\psi, \qquad A_2 = \sqrt{\frac{\pi}{2}}\, K_p \sin\delta. \tag{5}$$

The factor $\sqrt{\pi/2}$ undoes the normalisation of (4). A_1 and A_2 are the coordinates of S in the plane spanned by Φ_1 and Φ_2; the matched filter of Section **??** estimates them. The expansion of the norm that justifies keeping only the first-order term, and the resulting phase, are given in Appendix A.2.

Any channel that is a fixed linear combination of the cable field can be written as one vector of $\mathrm{span}\{\Phi_1, \Phi_2\}$:

$$s(\tau) = a \left(\cos\beta\, \Phi_1(\tau) + \sin\beta\, \Phi_2(\tau) \right), \qquad a \geq 0, \quad \beta \in (-180^\circ, 180^\circ]. \tag{6}$$

The amplitude a sets the signal level and the phase β sets the shape: $\beta = 0$ is the even bell Φ_1, $\beta = 90^\circ$ is the odd profile Φ_2, and intermediate phases are asymmetric (Figure 2c).

Hilbert relation. The observation underlying this paper is that Φ_2 is the Hilbert transform of Φ_1. The pair is therefore the real and imaginary parts of a single analytic template:

$$\Phi_2 = \mathcal{H}\{\Phi_1\}, \qquad \Psi(\tau) = \Phi_1(\tau) + i\,\Phi_2(\tau) = \sqrt{\frac{2}{\pi}} \frac{1}{1-i\tau}, \tag{7}$$

and the amplitude and phase of (6) are the modulus and argument of one complex coefficient:

$$s(\tau) = \Re\left\{ a\, e^{-i\beta}\, \Psi(\tau) \right\}. \tag{8}$$

This is the line-source instance of the classical property that the horizontal and vertical components of a 2D potential field form a Hilbert pair. Every channel the platform can form is thus one point $a\, e^{i\beta}$ of the complex plane, and the whole paper is an analysis of what each processing step does to that point.

Two properties of Ψ are used (Appendix A.4). Its spectrum is one-sided, so Φ_1 and Φ_2 have the same autocorrelation; the matched filter of Section **??** is then a quadrature receiver, whose output energy is the squared modulus of a single complex correlation and is therefore independent of β when the record is unbounded. And Ψ continues analytically off the real axis with a single pole at $\tau = -i$, which is what makes the coherence of a window with the span equal to one at the CPA and nowhere else (Appendix **??**). What truncation destroys is the first property, and the displacement it produces is read off the second harmonic $\sin 2\beta$ of the phase.

In the implementation, however, Φ_1 and Φ_2 are used as two real templates and no estimator manipulates complex quantities; the analytic form is a bookkeeping device for the amplitude and the phase.

Gradiometer channels. The gradiometer is the difference DUO − UNO. The lateral axis makes the angle θ with the cable normal, so the two sensors sample the same channel at normal coordinates that differ by $b\cos\theta$, i.e. by $h = b\cos\theta/d$ in τ. To first order in that difference (a central difference),

$$\begin{aligned} \Delta s(\tau) &\simeq \frac{b\cos\theta}{d}\,\frac{\mathrm{d}s}{\mathrm{d}\tau} = \frac{b\cos\theta}{\sqrt{2}\,d}\,a\,(\cos\beta\,\Gamma_1(\tau) + \sin\beta\,\Gamma_2(\tau))\,, \\ \Gamma_1 = \sqrt{2}\,\Phi_1' &= -\frac{4}{\sqrt{\pi}}\frac{\tau}{(1+\tau^2)^2}, \qquad \Gamma_2 = \sqrt{2}\,\Phi_2' = \frac{2}{\sqrt{\pi}}\frac{1-\tau^2}{(1+\tau^2)^2}. \end{aligned} \tag{9}$$

The factor $\sqrt{2}$ makes Γ_1 and Γ_2 unit-energy: with the integrals of Appendix A.3 and $(1-\tau^2)^2 = (1+\tau^2)^2 - 4\tau^2$,

$$\int \Phi_1'^2 = \frac{2}{\pi}\int \frac{4\tau^2}{(1+\tau^2)^4} = \frac{1}{2}, \qquad \int \Phi_2'^2 = \frac{2}{\pi}\left(\int \frac{\mathrm{d}\tau}{(1+\tau^2)^2} - 4\int \frac{\tau^2}{(1+\tau^2)^4}\right) = \frac{1}{2}.$$

Φ_1' is odd and Φ_2' is even, so Γ_1 and Γ_2 are orthogonal by parity (Figure 2b): the vertical gradiometer ΔB_z follows the even profile Γ_2 and the horizontal gradiometer follows the odd profile Γ_1. Since $B_y = 0$ everywhere, the horizontal cable field points along $\hat{x}$ at both sensors, and so does their difference: the horizontal gradiometer vector lies along the cable normal. The first-order model (9) follows from $s(\tau + h/2) - s(\tau - h/2) = h\,s'(\tau) + (h^3/24)\,s'''(\tau) + O(h^5)$, so its relative error is of order $h^2/24$ times the ratio of the third to the first derivative, with $h = 0.21$ at $d = 3$ m and $\theta = 50°$.

The gradiometer amplitude is reduced by $b|\cos\theta|/(\sqrt{2}\,d)$, 0.15 at $d = 3$ m and $\theta = 50°$, and it vanishes at perpendicular incidence. Two departures from (9) are kept in mind. Roll tilts the baseline, giving it the vertical component $b\sin(\text{roll})$; this mixes in the vertical derivative, an impurity of relative size about $\tan(\text{roll})/|\cos\theta|$, 8% at 3° of roll and 50° of incidence, oscillating at the wave period. Differences between the residual calibrations of the two sensors leave a differential term that also varies with attitude.

The channels of this paper have the following amplitudes and phases (with $K' = \sqrt{\pi/2}\,K_p$):

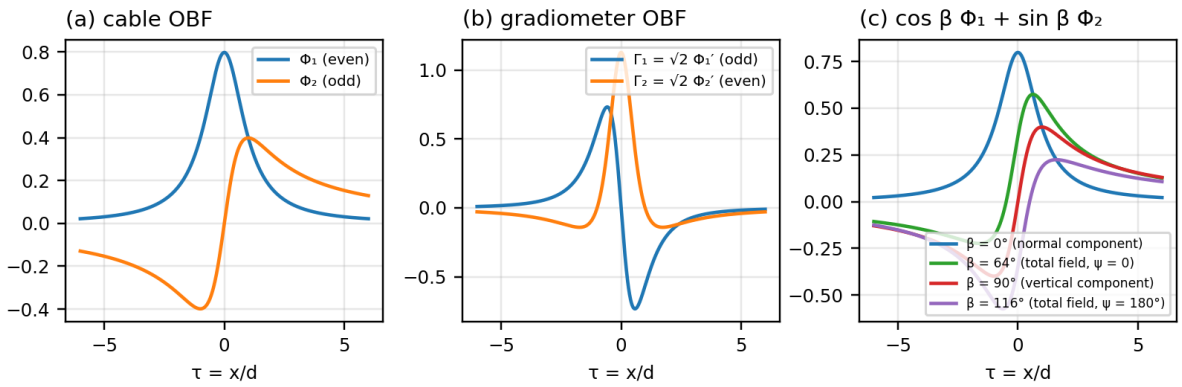


Figure 2: (a) Cable OBF: the even Φ_1 and the odd Φ_2 of (4). (b) Gradiometer OBF: the odd Γ_1 and the even Γ_2 of (9). (c) Channels of (6) for $\beta = 0°$ (normal component), 90° (vertical component), and the limits 64° and 116° of the total-field phase at Brest.

Table 1: Amplitude and phase of each channel in the basis $\{\Phi_1, \Phi_2\}$.

Channel	Amplitude a	Phase β
Normal component B_n	K'	0
Vertical component B_z	K'	90° (−90° for the opposite current)
North and East components, $B_n \cos\psi$ and $B_n \sin\psi$	$K'\lvert\cos\psi\rvert$, $K'\lvert\sin\psi\rvert$	0 or 180°
Total field S	$K'\sqrt{\cos^2\delta\,\cos^2\psi + \sin^2\delta}$	$\tan\beta = \tan\delta/\cos\psi$
Gradiometer, horizontal normal ΔB_n	$\dfrac{K'\,b\,\lvert\cos\theta\rvert}{\sqrt{2}\,d}$	0 on the gradiometer basis (odd profile Γ_1)
Gradiometer, vertical ΔB_z	$\dfrac{K'\,b\,\lvert\cos\theta\rvert}{\sqrt{2}\,d}$	90° on the gradiometer basis (even profile Γ_2)

The vector components have phases fixed by physics. The total-field phase follows from (5), $\tan\beta = A_2/A_1 = \tan\delta/\cos\psi$ (Appendix A.2). Since $\lvert\cos\psi\rvert \le 1$, we have $\lvert\tan\beta\rvert \ge \tan\delta$, so β lies between δ and $180° - \delta$: between 63.7° and 116.3° at Brest, for every cable bearing. The total field is therefore always closer to the odd profile than to the even one. At $\psi = 45°$, for example, $\beta = 70.7°$.

3 Detection and localization under truncation

The length of the matched-filter projection decides whether a cable is present; its position decides where. With an unbounded record both are maximal at the CPA, but on a finite pass only quantities that do not depend on that length still place the crossing correctly. This section shows why, and gives the localisers that follow from it.

3.1 Matched Filter and Detection

Let $m(\tau)$ be one recorded channel over a leg, expressed in the normalised coordinate 1.The matched filter correlates it with the two basis functions centred at the lag t (in CPA distances):

$$y_i(t) = \int m(\tau)\,\Phi_i(\tau - t)\,\mathrm{d}\tau, \quad i = 1, 2, \qquad \mathbf{y}(t) = \big(y_1(t),\, y_2(t)\big)^\top \tag{10}$$

In the implementation the integral is a sum over samples m_k taken at $\tau_k = k/(f_s T_s)$, with templates sampled on the same grid and normalised to unit Euclidean norm. For a channel 6 and an unbounded record, 10 is linear in $(\cos\beta, \sin\beta)$:

$$\mathbf{y}(t) = a\,\mathbf{C}(t)^\top \begin{pmatrix} \cos\beta \\ \sin\beta \end{pmatrix}, \qquad \mathbf{C}(t) = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}, \qquad c_{ij}(t) = \int \Phi_i(\tau)\,\Phi_j(\tau - t)\,\mathrm{d}\tau. \tag{11}$$

$c_{ij}(t)$ is the correlation of ϕ_i with ϕ_j shifted by t. The four functions have closed forms REF TO APPENDEX:

$$c_{11}(t) = c_{22}(t) = \frac{4}{4 + t^2}, \qquad c_{21}(t) = -c_{12}(t) = \frac{2t}{4 + t^2}. \tag{12}$$

The identity $c_{21} = -c_{12}$ follows from parity alone (Φ_1 even, Φ_2 odd). The identity $c_{22} = c_{11}$ is the consequence of the Hilbert relation announced in Section 2.2: Φ_1 and Φ_2 have the same autocorrelation. Together they make $\mathbf{C}(t)$ a rotation matrix multiplied by the scalar $2/\sqrt{4 + t^2}$, and a rotation does not change the length of the vector $(\cos\beta, \sin\beta)$. Hence

$$\|\mathbf{y}(t)\|^2 = y_1^2 + y_2^2 = \frac{4a^2}{4 + t^2} \qquad \text{for every phase } \beta. \tag{13}$$

The energy is maximal at $t = 0$, the CPA, falls to half at $|t| = 2$, and does not depend on which channel is used. With an unbounded record the energy peak therefore locates the crossing for every channel and every geometry.

Detection. The detector thresholds the energy normalised by the noise variance:

$$\Lambda(t) = \frac{\|\mathbf{y}(t)\|^2}{\sigma_n^2} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} \eta. \tag{14}$$

$\mathcal{H}_0$ is the hypothesis "no cable" and $\mathcal{H}_1$ "cable"; σ_n is the noise standard deviation per sample and η the threshold. Under white Gaussian noise, (**??**) is the GLRT for a signal in $\mathrm{span}\{\Phi_1, \Phi_2\}$ with unknown amplitudes reference my paper and other papers. With unit-norm, orthogonal templates, $\mathbf{y}$ is then Gaussian with covariance $\sigma_n^2\mathbf{I}$ under $\mathcal{H}_0$, so Λ follows a χ^2 law with 2 degrees of freedom and the false-alarm probability at one lag is $P_{\mathrm{fa}} = \exp(-\eta/2)$ ' (Appendix B.2) '. The noise of a USV is colored, and a leg contains many lags, so this analytical value does not hold in practice; the threshold is calibrated empirically instead.

3.2 Truncation Breaks the Phase Invariance

A leg restricts the record to $|\tau| \le K$. Two truncation models must be distinguished:

$$\begin{aligned} &\text{template truncation:} \quad y_i(t) = \int m(\tau)\,\Phi_i(\tau - t)\,\mathbf{1}_{|\tau - t| \le K}\,\mathrm{d}\tau; \\ &\text{record truncation:} \quad y_i(t) = \int m(\tau)\,\mathbf{1}_{|\tau| \le K}\,\Phi_i(\tau - t)\,\mathrm{d}\tau. \end{aligned} \tag{15}$$

1 is the indicator function. With template truncation the record is unbounded and the templates are clipped to K CPA distances about their centre. With record truncation the record is clipped to the leg and correlated with full templates. A survey leg is a record truncation, because the implementation correlates long templates ($|\tau| \le 30$) with whatever record exists.

In both models (11) still holds, with truncated correlations c_{ij}^K. Parity still gives $c_{12}^K(0) = c_{21}^K(0) = 0$, but the equal-autocorrelation identity fails. At $t = 0$ the two diagonal terms are the energies of Φ_1 and Φ_2 inside the pass:

$$E_1(K) = \frac{2}{\pi}\Big(\arctan K + \frac{K}{1+K^2}\Big), \qquad E_2(K) = \frac{2}{\pi}\Big(\arctan K - \frac{K}{1+K^2}\Big), \qquad E_1 - E_2 = \frac{4K}{\pi(1+K^2)}. \tag{15}$$

Φ_2 decays as $1/\tau$ while Φ_1 decays as $1/\tau^2$, so Φ_2 keeps more of its energy beyond the pass: $E_2 < E_1$. $\mathbf{C}^K(t)$ is no longer a scaled rotation, and the energy acquires a term proportional to $\sin 2\beta$ that is odd in t. Its slope at $t = 0$ moves the maximum. To first order in t (Appendix B.3) the energy peak sits at

$$\hat{\tau}_{\mathrm{EP}} \simeq \frac{4}{\pi K} \sin 2\beta, \qquad K \gg 1. \tag{16}$$

The displacement is driven by the energy imbalance $E_1 - E_2 \approx 4/(\pi K)$ between the truncated basis functions: it tilts the energy about the CPA, and the maximum moves towards the side the tilt favours, by an amount proportional to $\sin 2\beta$. Both truncation models share this law. The derivation, and its evaluation at finite K, are given in Appendix B.3.

Three properties follow. The displacement vanishes for the pure phases $\beta = 0°$ and $90°$, i.e. for the horizontal and vertical components. It changes sign with β, so it cannot be averaged out over passes of unknown phase. And it decays only as $1/K$: halving it requires doubling the pass.

Equation (16) is a first-order result. The exact displacement at $\beta = 45°$ exceeds it by about 30% at $K = 5$ (0.33 against 0.25 CPA distances), 15% at $K = 10$, 7% at $K = 20$ and 4% at $K = 30$ (Figure 3b). For short passes the higher-order terms dominate. Under record truncation, the survey case, the largest displacement over all phases and $K \ge 0.75$ is 0.65 CPA distances, at $K = 2$ and $\beta \approx 117°$. Within the total-field phase range at Brest (64–116°) it is 0.65, 0.28 and 0.12 CPA distances at $K = 2$, 5 and 10 (Figure 3c). At $d = 2$–4 m, 0.65 CPA distances is 1.3–2.6 m. Under template truncation the energy becomes bimodal as β approaches 90° at $K \approx 2$ (Figure 4a), so its maximum jumps between two lobes about two CPA distances apart (up to 2.1 in a ±3 search). Record truncation does not show this bimodality (Figure 4b).

The lawnmower lines of [6] operated at $K \gg 10$, where the displacement is small. A snake leg of amplitude 6 m over a standoff of 2–4 m operates at $K = 1.5$–3, where it is largest.

3.3 Parity Channels

References

[1] Khaled Hleis, Isabelle Quidu, Michel Legris, and Alain Bertholom. "Derivation of New Orthonormal Basis Functions for Magnetic Detection of Submarine Telecommunications Cables". In: *IET Radar, Sonar & Navigation* 20 (2026), e70225. DOI: 10.1049/rsn2.70225.

A Field of the Cable and the Channel Phases

A.1 Field of an Infinite Line Current

For an infinitely long straight conductor, the Biot–Savart law gives a field of magnitude $\mu_0 I/(2\pi r)$ at the distance r, directed along $\hat{I} \times \mathbf{r}/r$, where $\hat{I}$ is the unit vector of the current and $\mathbf{r}$ the vector from the conductor to the point, perpendicular to it. In the CPA frame the cable runs along $\hat{y}$ at $x = 0$, $z = d$, and the sensor is at $(x, \cdot, 0)$, so $\mathbf{r} = (x, 0, -d)$ and $r^2 = x^2 + d^2$. For a current along $-\hat{y}$, using $\hat{y} \times \hat{x} = -\hat{z}$ and $\hat{y} \times \hat{z} = \hat{x}$,

$$(-\hat{y}) \times (x\,\hat{x} - d\,\hat{z}) = d\,\hat{x} + x\,\hat{z}, \qquad \mathbf{B}_s = \frac{\mu_0 I}{2\pi(x^2+d^2)}\,(d,\,0,\,x) = \frac{\mu_0 I}{2\pi d}\left(\frac{1}{1+\tau^2},\,0,\,\frac{\tau}{1+\tau^2}\right), \tag{17}$$

with $\tau = x/d$, which is (3). The opposite current direction changes the sign of K_p. For a straight section of finite half-length ℓ on each side of the CPA, the field at the CPA is multiplied by $\ell/\sqrt{\ell^2 + d^2}$: at $\ell = 5d$ the infinite-line model is accurate to 2%.

A.2 Expansion of the Total Field and Its Phase

A scalar magnetometer, or the norm of a vector magnetometer, measures $|\mathbf{T} + \mathbf{B}_s|$, where $\mathbf{T}$ is the geomagnetic field. With $u = 2\,\mathbf{B}_s \cdot \hat{\mathbf{T}}/|\mathbf{T}| + |\mathbf{B}_s|^2/|\mathbf{T}|^2$ we have $|\mathbf{T} + \mathbf{B}_s| = |\mathbf{T}|\sqrt{1+u} = |\mathbf{T}|(1 + u/2 - u^2/8 + O(u^3))$, so, keeping terms up to second order in $|\mathbf{B}_s|/|\mathbf{T}|$,

$$S = |\mathbf{T} + \mathbf{B}_s| - |\mathbf{T}| = \mathbf{B}_s \cdot \hat{\mathbf{T}} + \frac{|\mathbf{B}_s|^2 - (\mathbf{B}_s \cdot \hat{\mathbf{T}})^2}{2|\mathbf{T}|} + O\!\left(\frac{|\mathbf{B}_s|^3}{|\mathbf{T}|^2}\right), \qquad \hat{\mathbf{T}} = \frac{\mathbf{T}}{|\mathbf{T}|}. \tag{18}$$

The second-order terms equal $|\mathbf{B}_{s\perp}|^2/(2|\mathbf{T}|)$, where $\mathbf{B}_{s\perp}$ is the part of $\mathbf{B}_s$ perpendicular to $\mathbf{T}$; they are at most $K_p^2/(2|\mathbf{T}|) = 100^2/(2 \times 47\,994) = 0.10$ nT at $d = 2$ m, i.e. 0.1% of the anomaly. Only the first term is kept in the analysis; all simulations use the exact norm.

The horizontal part of $\hat{\mathbf{T}}$ has magnitude $\cos\delta$ and points to magnetic north, and $\hat{x}$ makes the angle ψ with magnetic north, so $\hat{\mathbf{T}} \cdot \hat{x} = \cos\delta\cos\psi$ and $\hat{\mathbf{T}} \cdot \hat{z} = \sin\delta$ ($\hat{\mathbf{T}} \cdot \hat{y}$ is irrelevant because $B_y = 0$). With (17), the first-order term is

$$\mathbf{B}_s \cdot \hat{\mathbf{T}} = K_p\left(\frac{\cos\delta\cos\psi}{1+\tau^2} + \frac{\sin\delta\,\tau}{1+\tau^2}\right) = \sqrt{\frac{\pi}{2}}\,K_p\big(\cos\delta\cos\psi\,\Phi_1 + \sin\delta\,\Phi_2\big), \tag{19}$$

which is (5), since $1/(1+\tau^2) = \sqrt{\pi/2}\,\Phi_1$ and $\tau/(1+\tau^2) = \sqrt{\pi/2}\,\Phi_2$. Its amplitude and phase in

the sense of (6) are

$$a = \sqrt{A_1^2 + A_2^2} = \sqrt{\frac{\pi}{2}}\, K_p \sqrt{\cos^2\delta \cos^2\psi + \sin^2\delta}, \qquad \tan\beta = \frac{A_2}{A_1} = \frac{\tan\delta}{\cos\psi}, \tag{20}$$

with β in the quadrant of (A_1, A_2). Because $|\cos\psi| \le 1$, $|\tan\beta| \ge \tan\delta$, hence $\delta \le |\beta| \le 180^\circ - \delta$. At Brest $\tan\delta = 43\,023/21\,271 = 2.023$ ($\delta = 63.7^\circ$); for $\psi = 45^\circ$, $\tan\beta = 2.023/0.707 = 2.86$, so $\beta = 70.7^\circ$ and $a = 0.95\,\sqrt{\pi/2}\,K_p$.

A.3 Standard Integrals and Orthonormality

The integrals used throughout are

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}\tau}{(1+\tau^2)^2} = \frac{\pi}{2}, \qquad \int_{-\infty}^{\infty} \frac{\mathrm{d}\tau}{(1+\tau^2)^3} = \frac{3\pi}{8}, \qquad \int_{-\infty}^{\infty} \frac{\tau^2\,\mathrm{d}\tau}{(1+\tau^2)^4} = \frac{\pi}{16}. \tag{21}$$

Thus $\int \Phi_1^2 = (2/\pi)(\pi/2) = 1$, and since $\tau^2/(1+\tau^2)^2 = 1/(1+\tau^2) - 1/(1+\tau^2)^2$, $\int \Phi_2^2 = (2/\pi)(\pi - \pi/2) = 1$. The product $\Phi_1\Phi_2$ is odd, so $\int \Phi_1\Phi_2 = 0$: the pair (4) is orthonormal over the real line. The last two integrals of (21) give the unit energy of the gradiometer basis (9).

A.4 The Analytic Template: Hilbert Pair, Spectrum and Poles

With the Fourier convention $\hat{f}(\omega) = \int f(\tau)\, e^{-i\omega\tau}\,\mathrm{d}\tau$, the transform of $1/(1+\tau^2)$ is $\pi e^{-|\omega|}$, so

$$\hat{\Phi}_1(\omega) = \sqrt{2\pi}\, e^{-|\omega|}, \qquad \hat{\Phi}_2(\omega) = -i\,\mathrm{sgn}(\omega)\,\hat{\Phi}_1(\omega), \tag{22}$$

the second because the Hilbert transform multiplies a spectrum by $-i\,\mathrm{sgn}\,\omega$, and the inverse transform of $-i\pi\,\mathrm{sgn}(\omega)\, e^{-|\omega|}$ is $\tau/(1+\tau^2)$; hence $\Phi_2 = \mathcal{H}\{\Phi_1\}$. Two consequences follow.

First, $|\hat{\Phi}_2| = |\hat{\Phi}_1|$, so by the Wiener–Khinchin theorem Φ_1 and Φ_2 have the same autocorrelation function. Equivalently, the analytic template $\Psi = \Phi_1 + i\Phi_2$ of (7) has the one-sided spectrum

$$\hat{\Psi}(\omega) = (1 + \mathrm{sgn}\,\omega)\hat{\Phi}_1(\omega) = \begin{cases} 2\sqrt{2\pi}\, e^{-\omega}, & \omega > 0, \\ 0, & \omega < 0, \end{cases} \qquad \int \Psi(\tau)\,\Psi^*(\tau - t)\,\mathrm{d}\tau = \frac{4}{2 - it}. \tag{23}$$

Correlating a channel with Ψ is therefore a quadrature receiver, and the modulus of (23), $4/\sqrt{4+t^2}$, carries no dependence on the phase β of (8): this is the phase invariance exploited in Section **??**, and what record truncation destroys.

Second, $\Psi(\tau) = \sqrt{2/\pi}/(1 - i\tau)$ continues analytically off the real axis as a rational function with the single pole $\tau = -i$. A shifted template $\Psi(\tau - t)$ has its pole at $t - i$, and two rational functions that agree on an interval agree everywhere, so their poles must coincide: for real t this forces $t = 0$. This is the argument used in Appendix **??** to show that the subspace coherence of a window with $\mathrm{span}\{\Phi_1, \Phi_2\}$ reaches one at the CPA and nowhere else, and that a wrong template scale λ moves the pole to $t - i\lambda$ and so prevents it.